

DRY-RUN GUIDE | alternate_positive_negative.py
Source: Python\3rd_Belt\alternate_positive_negative.py

CORE BEHAVIOR

Rearranges positive and negative values in alternating positions using the source's chosen strategy.

REPRESENTATIVE INPUT

'[1,2,-3,-4,5,-6]'.

STEP TRACE

Separate/locate signs, then place one positive and one negative in successive slots.

EXPECTED OUTPUT / RETURN

A possible order: '[1,-3,2,-4,5,-6]'; exact order depends on implementation.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt\alternate_positive_negative.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Arrange the numbers so positive and negative values alternate in order.
002 | # Add your solution here.
003 |
004 | def rearrange(arr):
005 |     # Separate positive (including 0) and negative numbers
006 |     pos = [x for x in arr if x >= 0]
007 |     neg = [x for x in arr if x < 0]
008 |
009 |     # If all elements belong to only one category, preserve original order
010 |     if not pos or not neg:
011 |         return arr
012 |
013 |     result = []
014 |     i, j = 0, 0
015 |
016 |     # Alternate between positive and negative numbers
017 |     while i < len(pos) and j < len(neg):
018 |         result.append(pos[i])
019 |         result.append(neg[j])
020 |         i += 1
021 |         j += 1
022 |
023 |     # Append any remaining elements to the end
024 |     if i < len(pos):
025 |         result.extend(pos[i:])
026 |     if j < len(neg):
027 |         result.extend(neg[j:])
028 |
029 |     return result
030 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.